

1. What is a media converter, and why is it used?
2. Differentiate between a hub and a switch.
3. Explain the function of a repeater.
4. Define network topology and describe the most common types, such as bus, star, and ring.
5. Discuss the advantages and limitations of bus topology.
6. Compare star and hybrid topologies.
7. Explain the difference between LAN and WAN.
8. Describe the uses of MAN.
9. Explain PAN with an example.
10. What is network architecture, and why is it important in network design?
11. Compare and contrast the client-server and peer-to-peer network architectures.
12. Explain IPv4 with an example.
13. Describe IPv6 and its advantages.
14. What is a MAC address?
15. Define an Intranet and an Extranet, and highlight their differences.
16. Write about any four services of the Internet in your daily life.
17. What is a network protocol, and why is it essential for Communication?

3. Write the technical term for the following:

No.	Statement	Technical Term
a	Transmission of text, audio, and video through some electronic means.	Telecommunication
b	Data transmission is only in one direction.	Simplex
c	Transmission of data both ways, but one at a time.	Half-Duplex
d	Transmission of data in both directions simultaneously.	Full-Duplex
e	An interconnection between two or more computers.	Computer Network
f	A wire made of copper twisted together and insulated.	Twisted Pair Cable
g	Cable with a central conductor, insulating layer, and shield.	Coaxial Cable
h	A cable that carries data as light waves.	Fiber Optic Cable
i	Radio waves for audio transmission.	Radio Communication
j	High-frequency waves for long-distance communication.	Microwaves
k	Provides a port to connect a computer to a network.	Network Interface Card (NIC)
l	A device used to connect a PC to a telephone line.	Modem
m	The computer on which users run applications.	Client

No.	Statement	Technical Term
n	Each computer or device on a network.	Node
o	A device that controls two dissimilar networks.	Gateway
p	A computer that provides services to other computers.	Server
q	An operating system that can handle a network.	Network Operating System (NOS)
r	An important software that controls and manages computer networks.	Network Operating System (NOS)
s	Physical structure of a computer network.	Network Layout
t	An interconnected pattern of network components.	Network Topology
u	The smallest type of computer network.	PAN (Personal Area Network)
v	Largest computer network.	WAN (Wide Area Network)
w	Privately owned network of computers.	Intranet
x	Network of networks.	Internet
y	Sending and receiving messages electronically via the Internet.	E-mail (Electronic Mail)

4. Write the full form of:

No.	Abbreviation	Full Form
a	LAN	Local Area Network
b	WAN	Wide Area Network
c	MAN	Metropolitan Area Network
d	PAN	Personal Area Network
e	UTP	Unshielded Twisted Pair
f	STP	Shielded Twisted Pair
g	BNC	Bayonet Neill–Concelman
h	RJ-45	Registered Jack-45
i	NIC	Network Interface Card
j	ISP	Internet Service Provider
k	WWW	World Wide Web
l	URL	Uniform Resource Locator
m	SMTP	Simple Mail Transfer Protocol

No.	Abbreviation	Full Form
n	FTP	File Transfer Protocol
o	HTTP	HyperText Transfer Protocol
p	TCP/IP	Transmission Control Protocol / Internet Protocol
q	VoIP	Voice over Internet Protocol
r	DNS	Domain Name System
s	DSL	Digital Subscriber Line
t	ADSL	Asymmetric Digital Subscriber Line
u	IPX/SPX	Internetwork Packet Exchange / Sequenced Packet Exchange
v	ARPANET	Advanced Research Projects Agency Network
w	POP	Post Office Protocol
x	MAC	Media Access Control
y	ICT	Information and Communication Technology
z	Email	Electronic Mail
aa	E-fax	Electronic Fax
ab	ATM	Asynchronous Transfer Mode

5. Multiple Choice Questions

1. Telecommunications mainly deals with:

- a. Storing data in computers
- b. Repairing hardware
- c. Sending and receiving data over long distances
- d. Designing software

2. A computer network mainly helps computers to:

- a. Increase screen size
- b. Improve graphics
- c. Connect and share resources

d. Reduce electricity use

3. Data packets are:

a. Large storage devices

b. Antivirus programs

c. Small pieces of data sent over a network

d. Hardware chips

4. Frequency is measured in:

a. Volt

b. Watt

c. Hertz

d. Byte

5. Broadband refers to:

a. Slow dial-up connection

b. Cable TV only

c. High-speed Internet connection

d. Satellite TV only

6. Bandwidth is:

a. Actual data speed

b. Delay in transmission

c. Maximum data capacity of a network

d. Variation in signal

7. Latency is measured in:

a. Seconds only

b. Bytes

c. Milliseconds

d. Hertz

8. Compared to 3G, 4G provides:

- a. Slower speed
- b. Same speed
- c. Faster data speed
- d. No internet access

9. 5G technology offers:

- a. Lower speed than 4G
- b. Only voice calls
- c. Very high speed and low latency
- d. Wired connectivity only

10. Coaxial cable has:

- a. Only one conductor
- b. Two conductors
- c. No shielding
- d. Plastic wires only

11. Fiber optic cable transmits data using:

- a. Electrical signals
- b. Magnetic waves
- c. Copper wires
- d. Light signals

12. Bluetooth is an example of:

- a. Broadcast radio
- b. Cellular radio
- c. Infrared
- d. Satellite

13. RFID uses:

- a. Infrared light
- b. Copper wire
- c. Optical fiber
- d. Radio waves

14. RJ45 connector is commonly used with:

- a. Fiber optic cable
- b. Coaxial cable
- c. Twisted pair cable
- d. Satellite dish

15. A media converter works at:

- a. Physical layer
- b. Network layer
- c. Application layer
- d. Transport layer

16. A switch mainly uses:

- a. IP address
- b. MAC address
- c. DNS
- d. URL

17. In bus topology, a terminator is used to:

- a. Increase speed
- b. Boost signal
- c. Stop signal reflection
- d. Encrypt data

18. Ring topology uses:

- a. Token passing
- b. Broadcasting
- c. Random access
- d. Cloud storage

19. In a star topology, all nodes connect to:

- a. Each other directly
- b. A central device
- c. Backbone cable only
- d. Token server

20. PAN covers:

- a. Entire country
- b. Whole city
- c. Short personal range
- d. Multiple continents

21. MAN covers:

- a. Personal room
- b. Single building
- c. Entire globe
- d. City area

22. In P2P architecture:

- a. A central server is required
- b. All computers are equal
- c. Only one computer shares
- d. Only LAN allowed

23. Which of the following is a valid example of an IPv4 address?

- a. 2001:0db8:85a3::8a2e:0370:7334
- b. 192.168.1.1
- c. FE80::0202:B3FF:FE1E:8329
- d. 1200:0000:AB00:1234

24. Which of the following is a valid example of an IPv6 address?

- a. 172.16.0.1
- b. 10.0.0.256
- c. 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- d. 255.255.255.0

25. Ping command is used to:

- a. Check connectivity
- b. Format disk
- c. Install OS
- d. Encrypt data

26. An intranet is:

- a. Public global network
- b. Open social media
- c. Private organizational network
- d. Satellite system

27. An intranet is mainly used within:

- a. A single organization
- b. The entire world
- c. Satellite systems
- d. Mobile towers

28. A communication protocol defines:

- a. Hardware price
- b. Cable color
- c. Screen resolution
- d. Rules for data exchange

29. Which of the following best describes an extranet?

- a. A public social media site
- b. A private network accessible only inside one office
- c. A private network that gives limited access to outsiders
- d. A wireless personal network